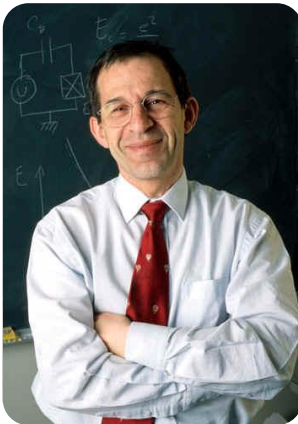


Superconducting Whispering Gallery Mode Resonators

Zlatko Minev



Michel H. Devoret



Ioan Pop



Dominic Kwok

Thanks to:

Kurtis

Nick

Matt R.

Nissim

Baleegh

Teresa

C. Pang

Luigi

R. Schoelkopf

QuLab

RSL Lab

Motivation

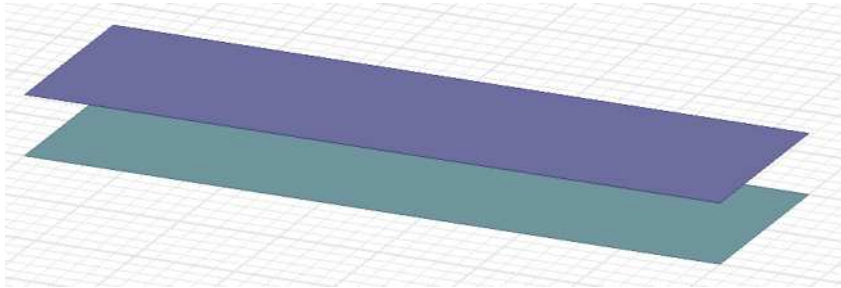
- I. `Wafer-Scalable` Circuit-QED architecture
 - A. Multiple resonators & qubits
 - B. Allow flux bias for a qubit coupled to a high-Q superconducting microwave cavity

- II. What is the intrinsic limit to superconducting devices?
 - I. Use a simple and robust platform to characterize cryogenic material properties of metals and dielectrics at the single photon level

Part I: `Wafer-Scalable` Circuit-QED architecture

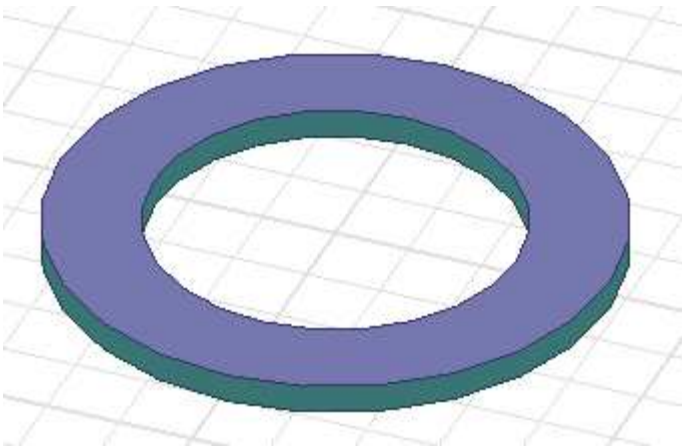
- 1. Concept: Photon on a Ring & Modular Setup**
- 2. Experimental Setup**
- 3. Characterization:**
 - 1. Modes & Quality factors**
 - 2. Lifetimes & Coherence Times**

Concept I: Photon Particle on a Ring

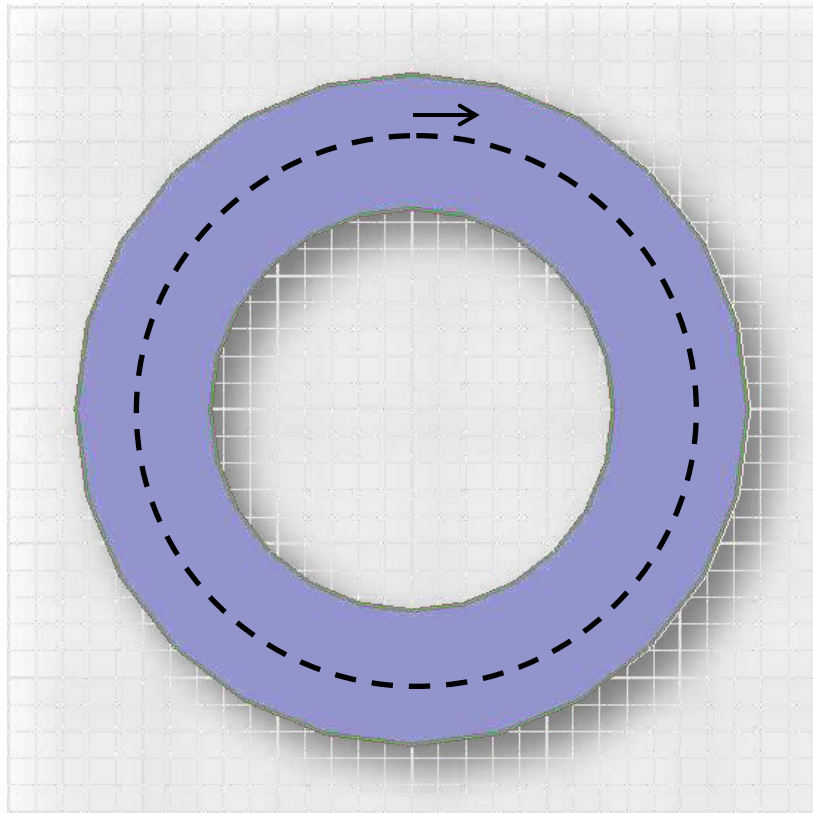
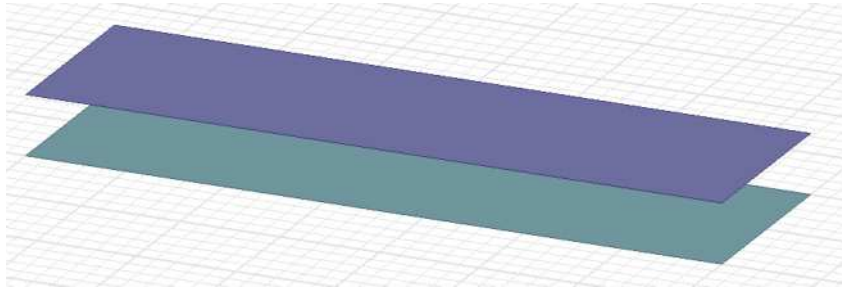


Design Strategy:

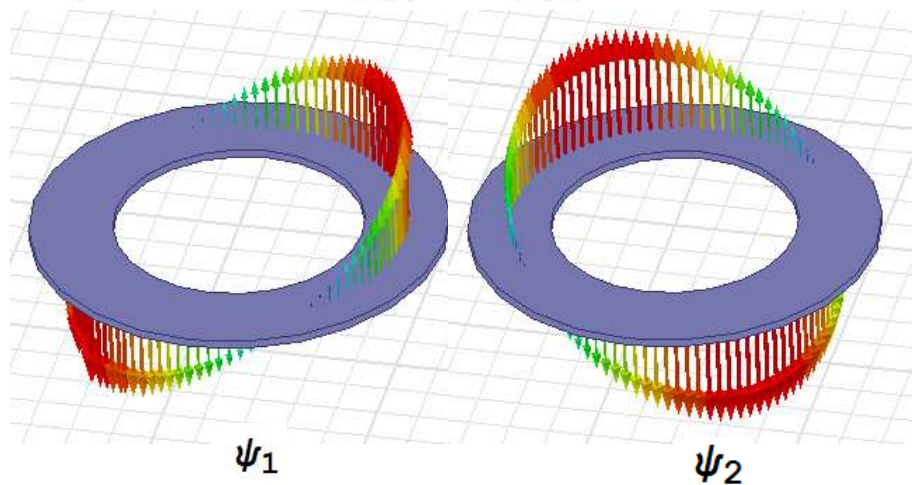
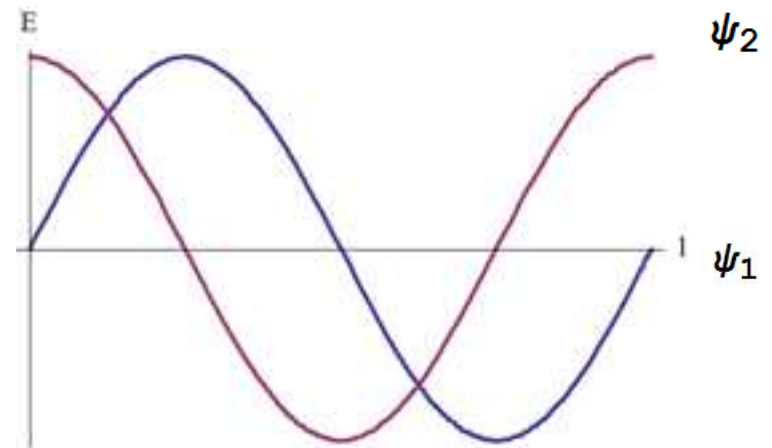
- Confine energy in lossless vacuum
- Keep geometry & design parameters simple



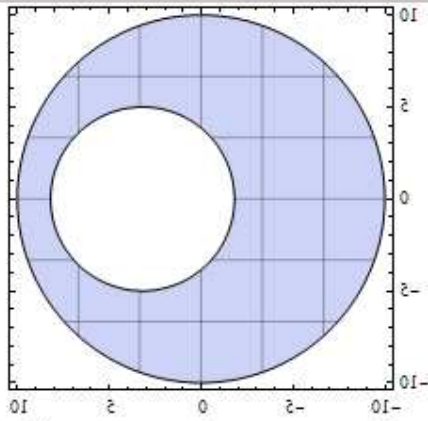
1D Description – 2 Degenerate Ground Modes



$$\nabla^2 \psi + \frac{w^2}{c^2} (1 + \Delta n(x)^2) \psi = 0$$

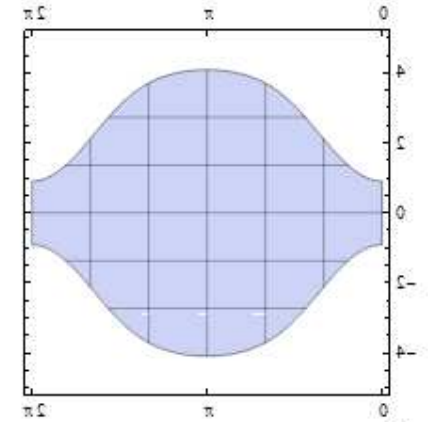


Lift Degeneracy

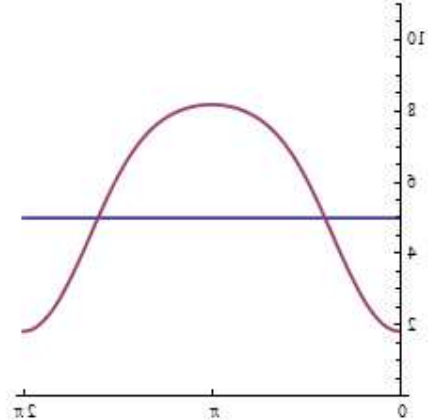


Simplified impedance
view from top

$$Z \propto \sqrt{\frac{L}{C}}$$

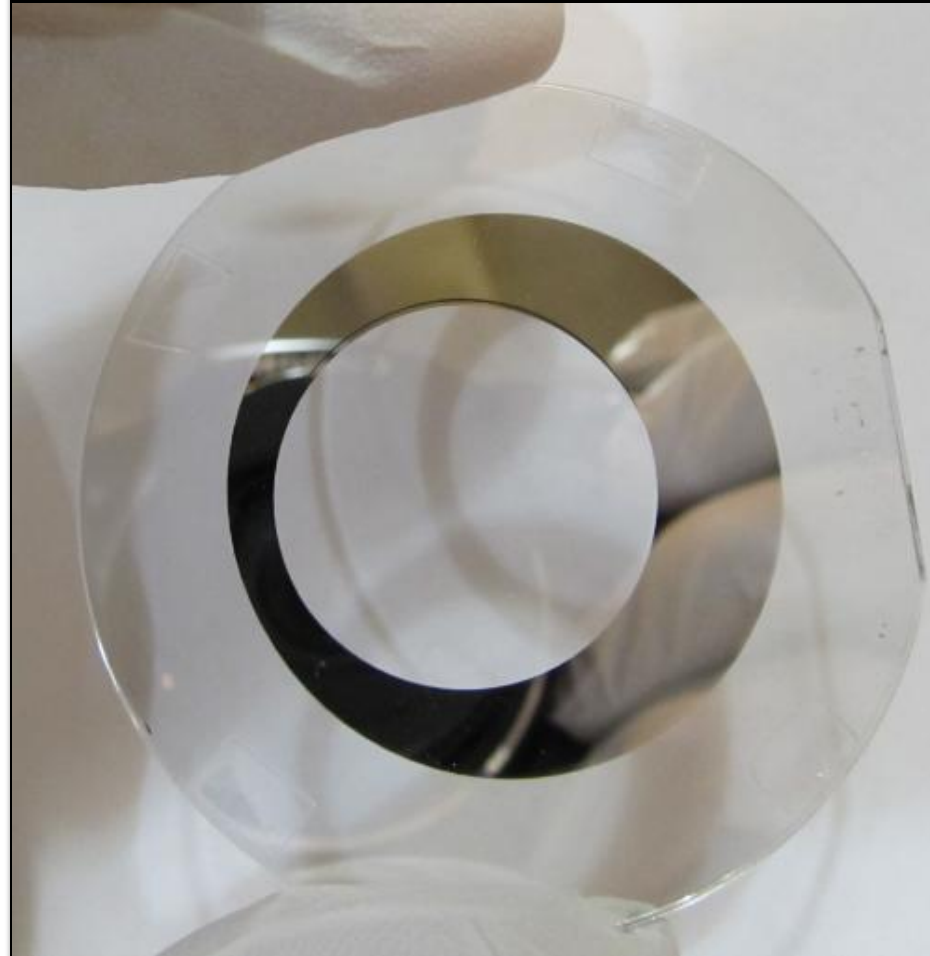


What the Photon
sees along its path
of travel

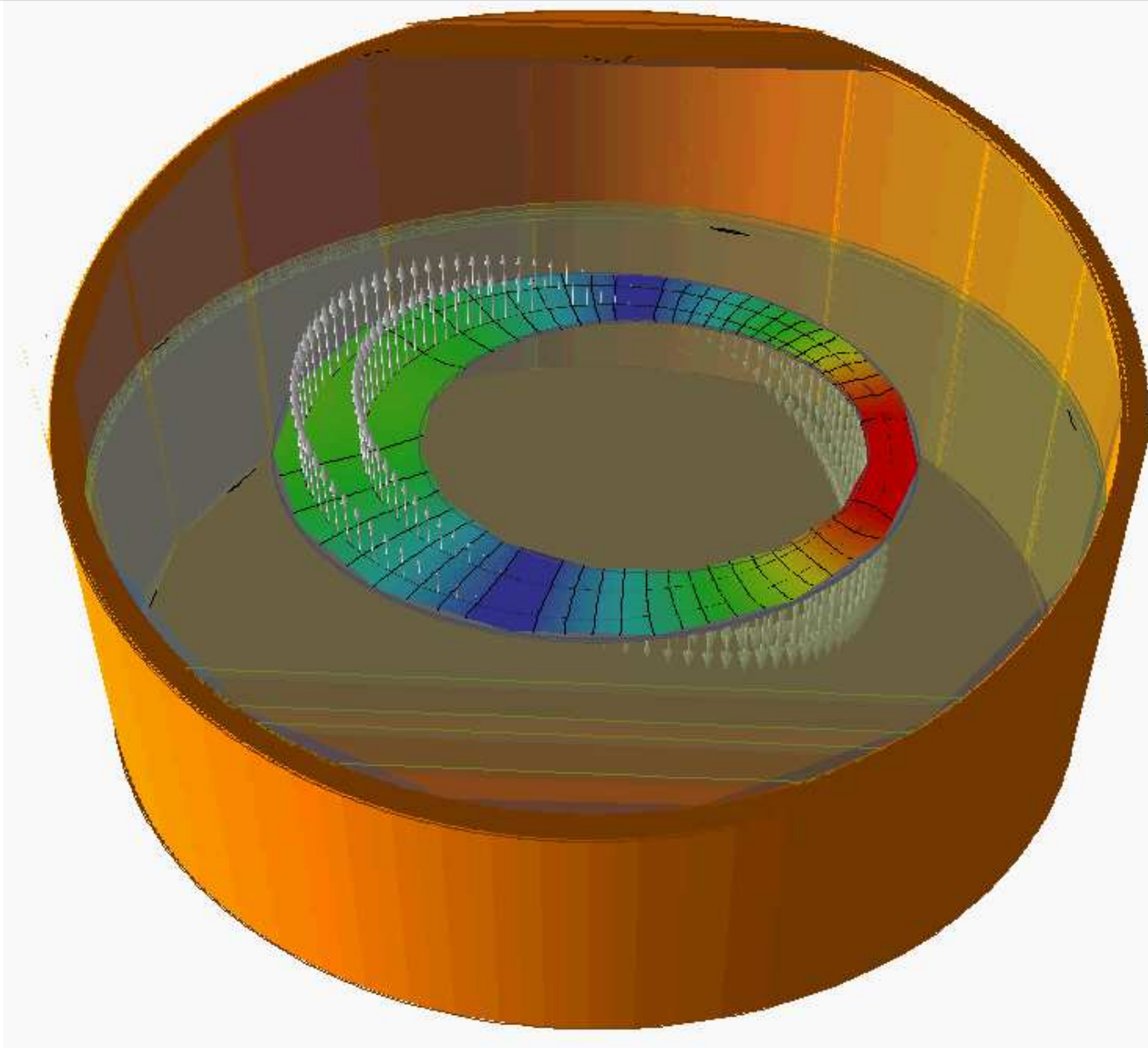


An effective 1D
admittance
for the photon along
its path

$$Y_{\infty \text{ TL}} = \frac{1}{377 \Omega} \frac{w}{d}$$

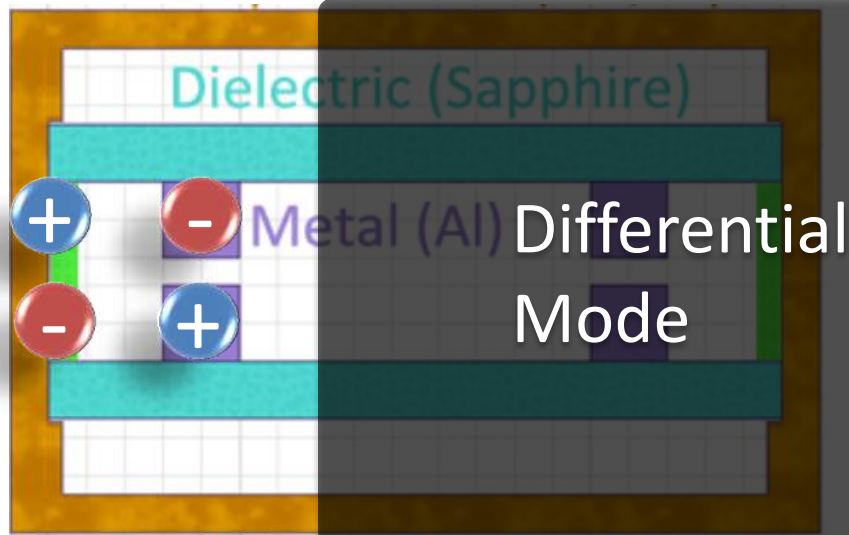


Differential Parallel Mode: Electric Fields



Concept II: Modular Setup

Test Metal



Test Dielectric

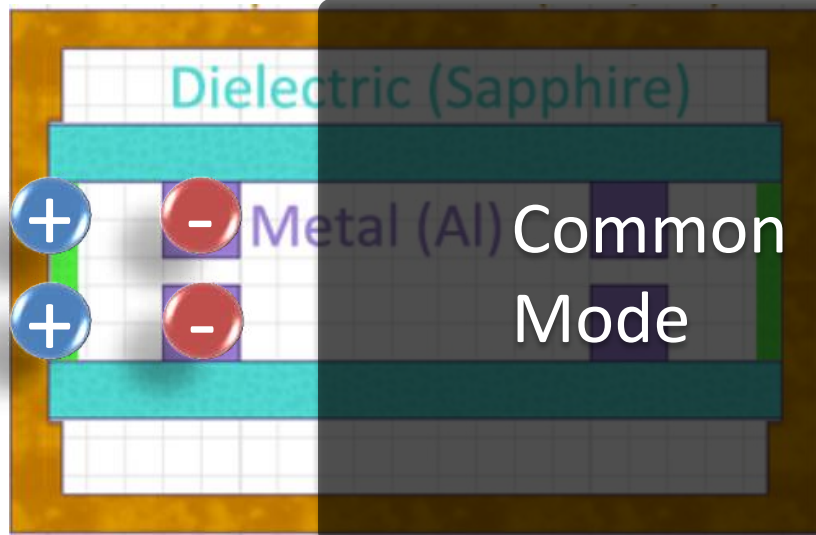


Not To Scale!

- Experimental setup is modular
- In the `test metal` setup, the thin-film rings confine the fields in lossless vacuum
- In the `test dielectric` setup, the fields are confined in the substrate dielectric

Concept II: Modular Setup

Test Metal



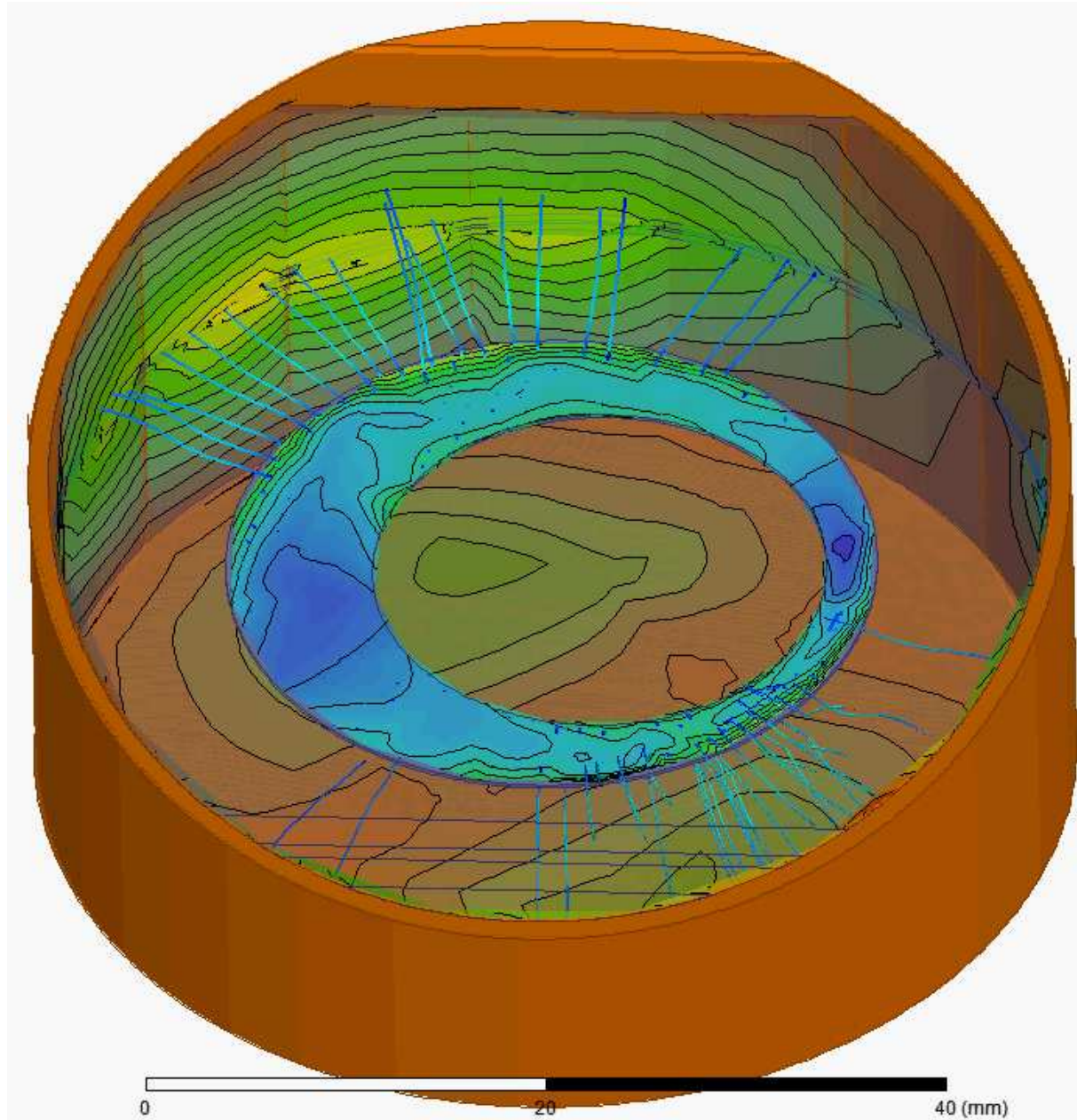
Test Dielectric



Not To Scale!

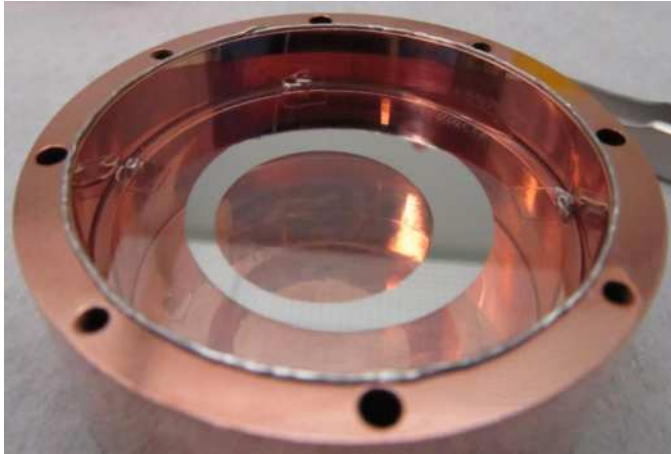
- Experimental setup is modular
- In the `test metal` setup, the thin-film rings confine the fields in lossless vacuum
- In the `test dielectric` setup, the fields are confined in the substrate dielectric

Common Perpendicular Mode: Electric Fields

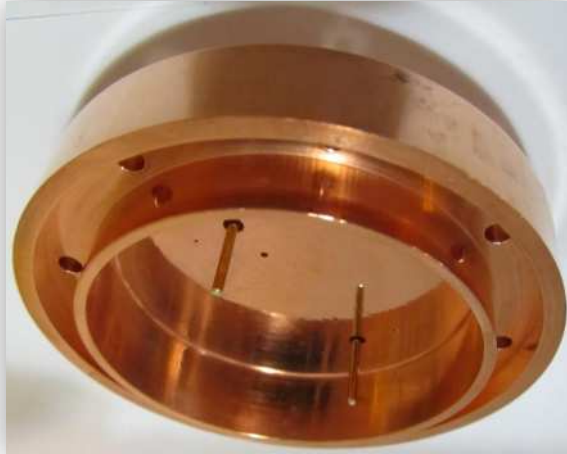


Modular Cavity & Sample Holder

Sample Holder Bottom



Sample Holder Top



SMA Pin Coupler
(Non Magnetic)

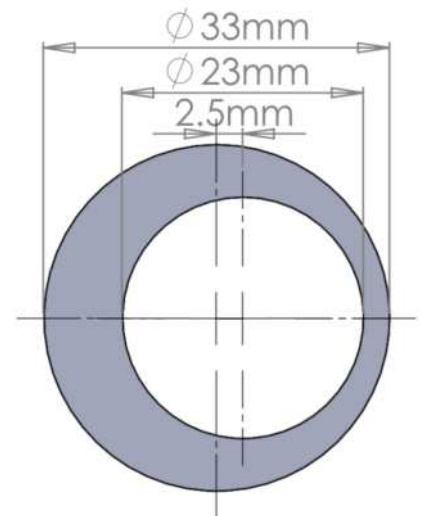


- Sample holder can be copper or aluminum
- Use custom *non-magnetic* SMA pin coupler

Technical Parameters:

- Al film thickness 300 nm
- O₂ plasma cleaning prior to deposition
- Sapphire wafer thickness 450 μm
- Wafer spacing 200 μm

Ring Dimensions



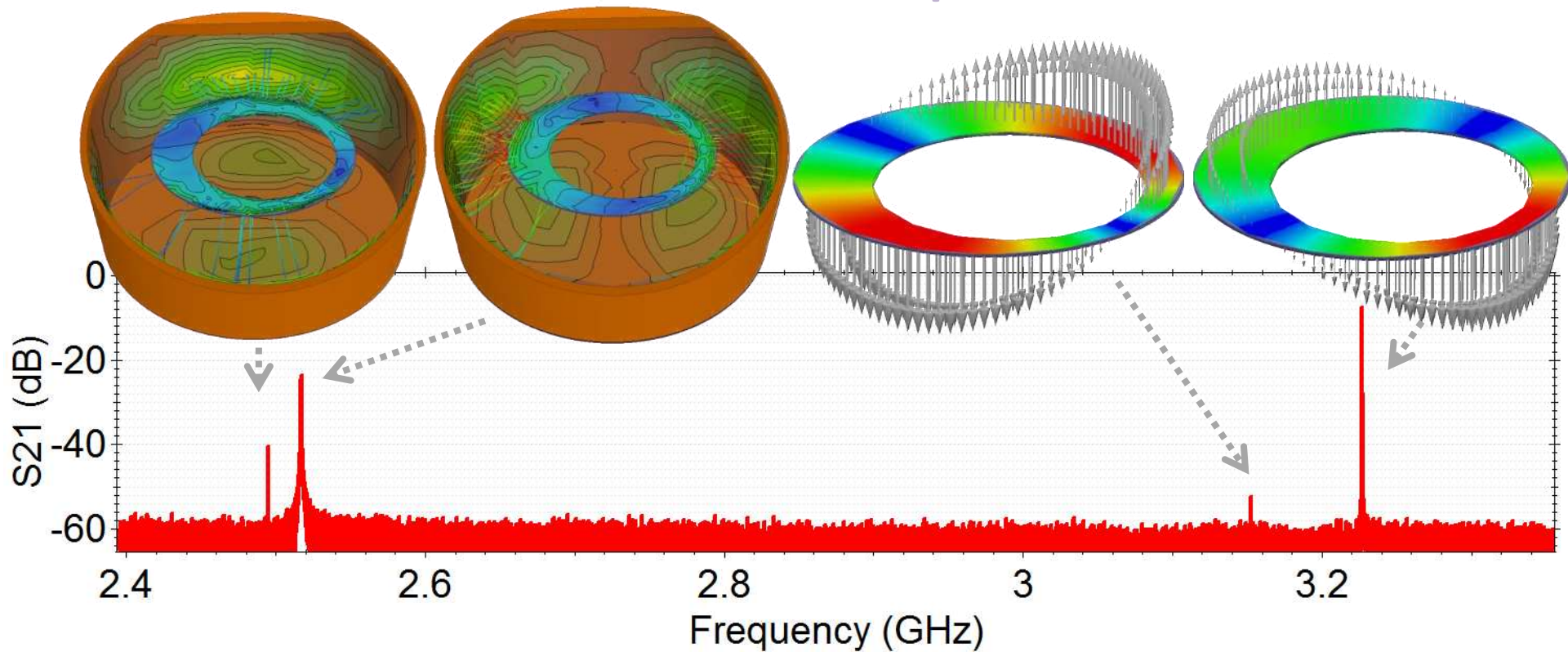
Identifying Modes (S_{21})

Common
Perpendicular

Common
Parallel

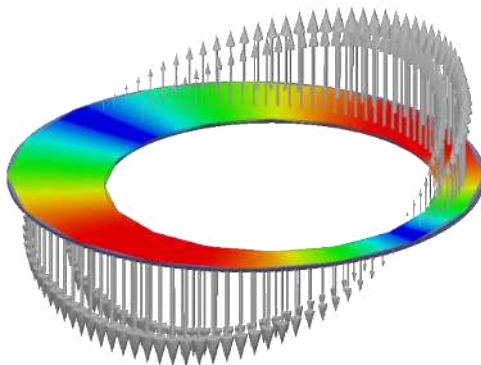
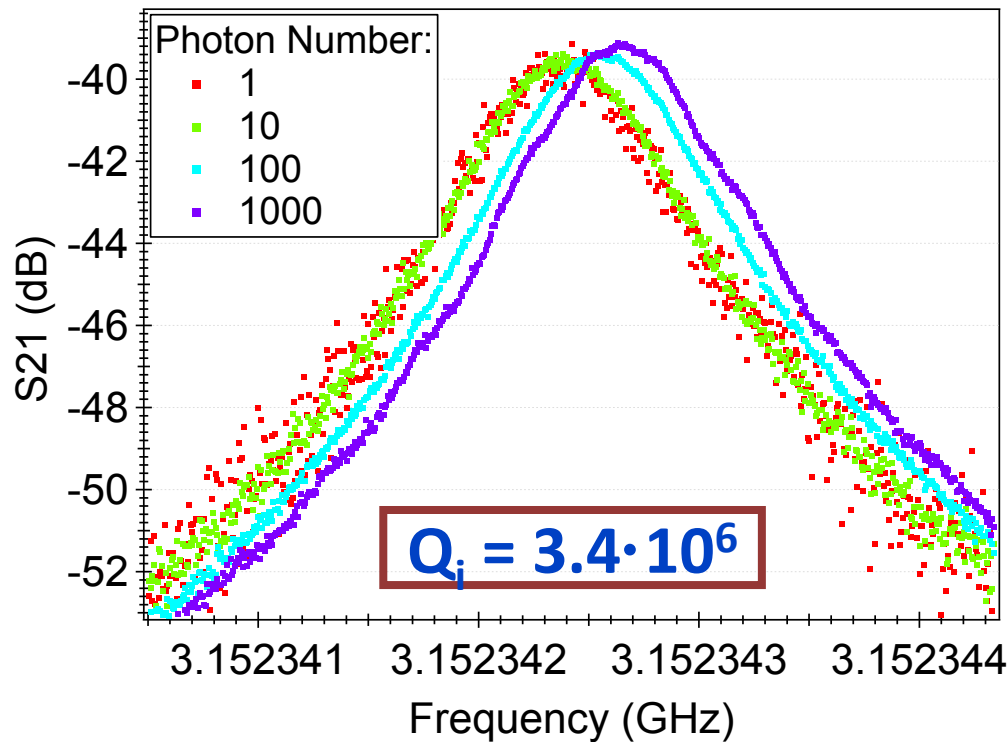
Differential
Perpendicular

Differential
Parallel



- The modes are easily identified and understood with the help of HFSS numerical simulations (Top Picture)
- Differential modes contain > 98% of mode energy in between the rings

Total Q vs Power (S_{21})



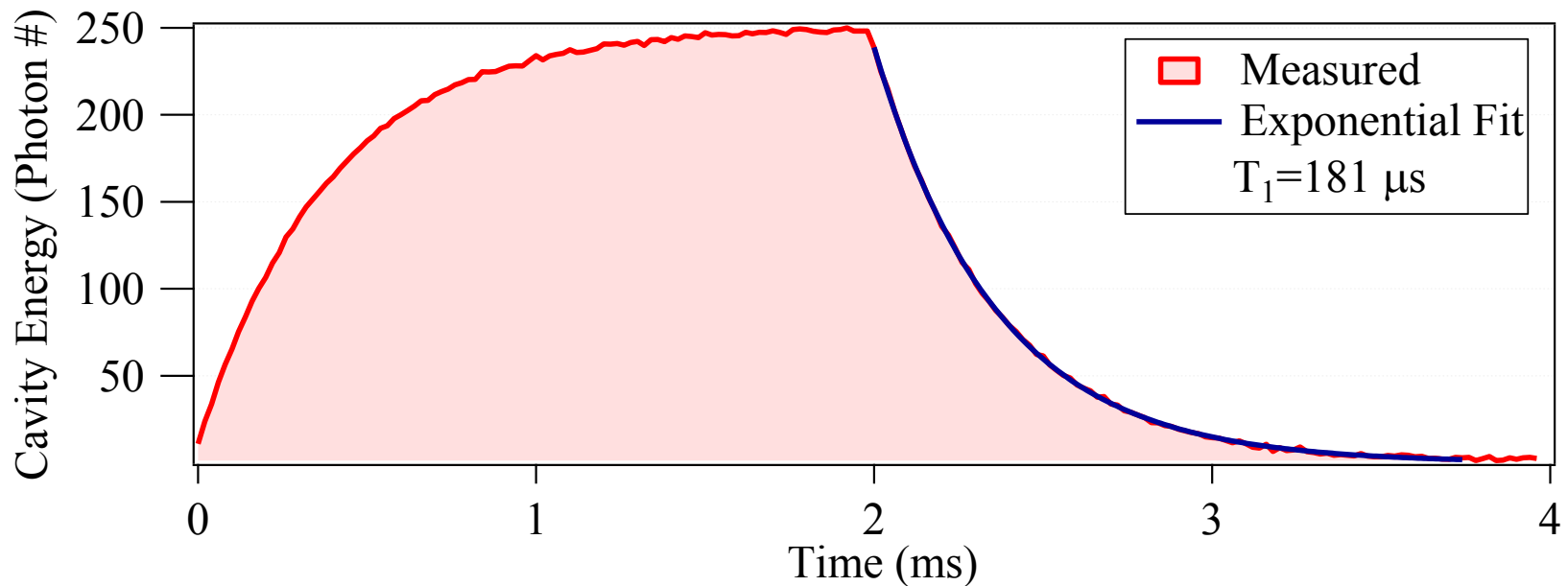
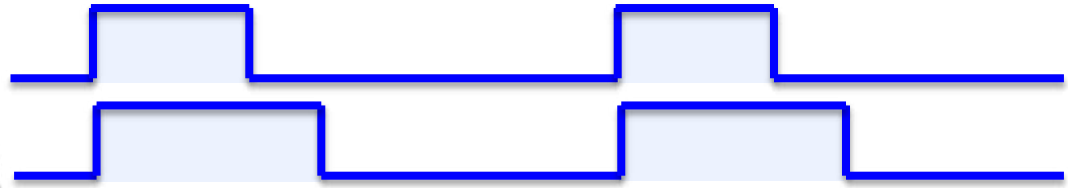
- Differential perpendicular mode in `test metal` configuration
- $BW = 940 \text{ Hz}$
- Q_i is power independent
- Undercoupled
- Single Photon Current = 80 nA
- Aluminum sample holder
- Copper sample holder measured $Q = 2 \cdot 10^5$

Energy Relaxation (T_1)

Pulse
Sequence

Pulse

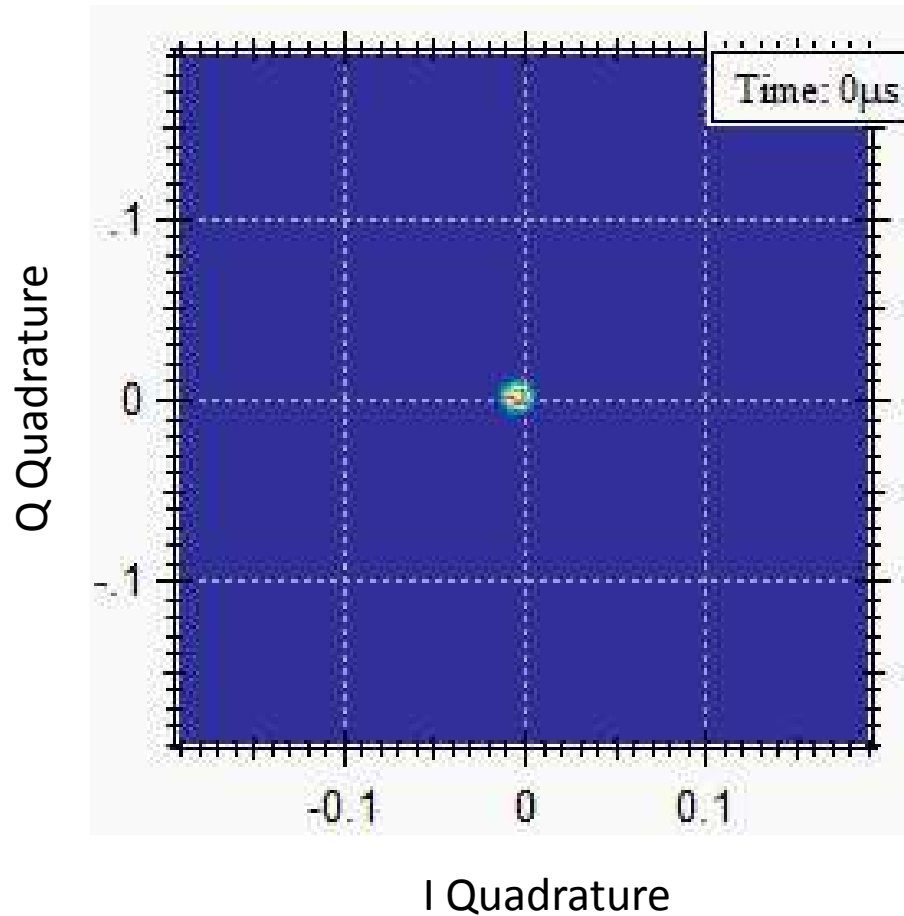
Measure



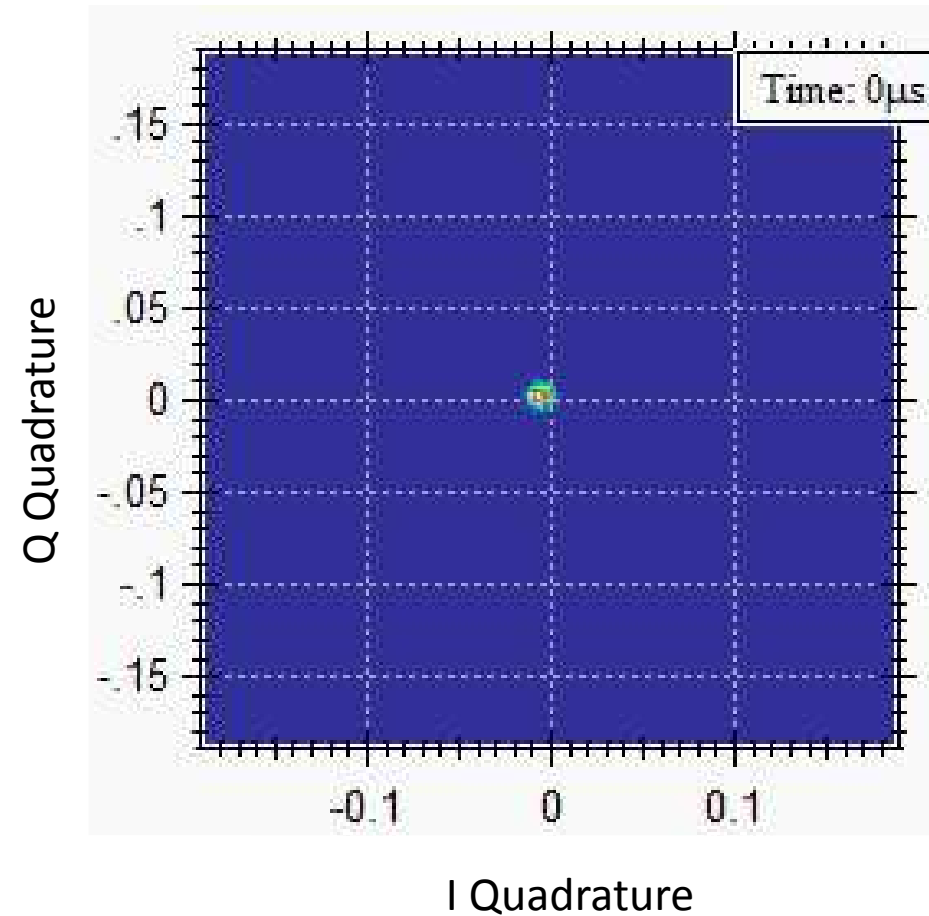
Heterodyne measurement of ring-up and ring-down of differential perpendicular mode in `test metal` set-up

In IQ Space: Overcoupled Mode & Mechanical noise

Roots on & Dephasing



Roots off

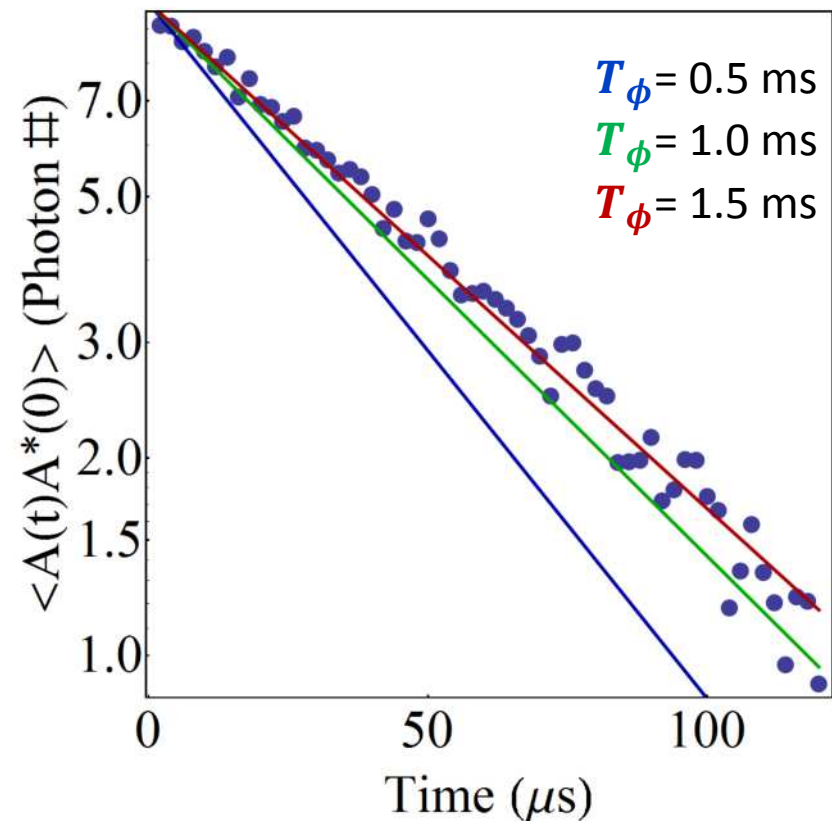


Phase Coherence (T_ϕ)

- Free relaxation autocorrelation function measures T_2
- $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}$
- Use circular statistics and wrapped normal distribution
- Assume uncorrelated phase and magnitude noise

$$\begin{aligned}
 & \langle A(t+\tau)A^*(t) \rangle \\
 = & \langle V(t+\tau)V(t) \rangle \langle e^{i(\phi(t+\tau)-\phi(t))} \rangle \\
 = & V_0^2 e^{-\frac{1}{2T_1}\tau} \langle e^{i\phi(\tau)} \rangle \\
 = & V_0^2 e^{-\frac{1}{2T_1}\tau} e^{-\frac{1}{T_\phi}\tau} \\
 = & V_0^2 e^{-\frac{1}{T_2}\tau}
 \end{aligned}$$

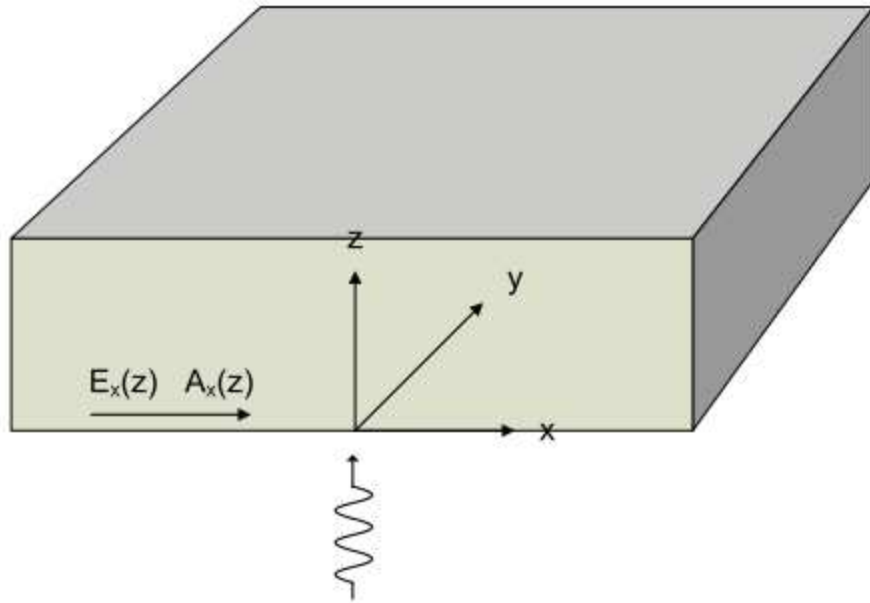
$$A = I + iQ$$



Part II: On the `Dissipation Budget`

- 1. Surface impedance & surface Q**
- 2. Aluminum thin film measurements**
- 3. Sapphire loss tangent measurements**

Surface Impedance



$$Z_S = \frac{E_{\text{tangential}}(x)}{H_{\text{tangential}}(y)} \Big|_{\text{at surface } (z=0)}$$
$$= R_s + i X_s$$

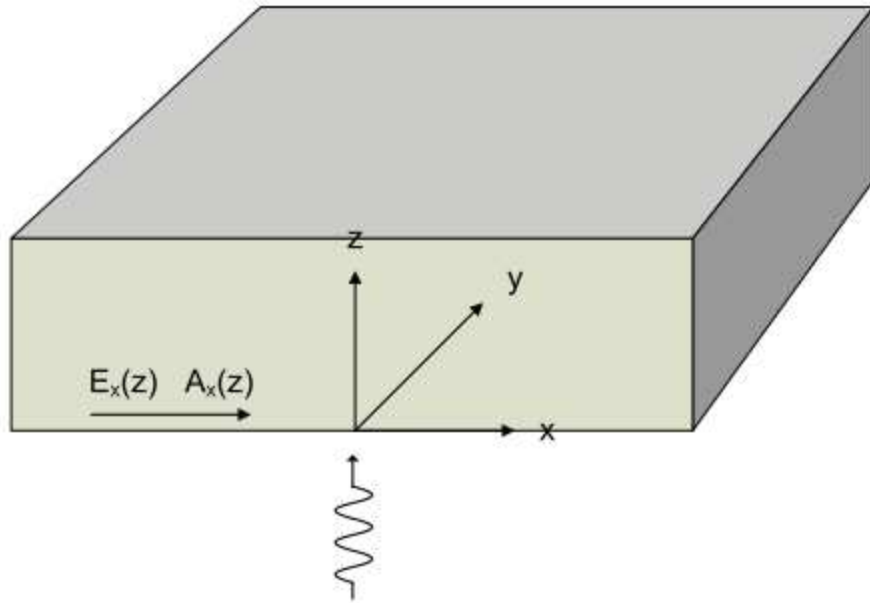
Rate of energy
dissipation in δ

Quasiparticles

$\sim$ EM energy
stored in δ

Kinetic
Inductance

Surface Impedance & Surface Q



$$Z_S = \frac{E_{\text{tangential}}(x)}{H_{\text{tangential}}(y)} \Big|_{\text{at surface } (z=0)}$$
$$= R_s + i X_s$$

Rate of energy
dissipation in δ

$\sim$ EM energy
stored in δ

$$Q_s = X_s / R_s$$

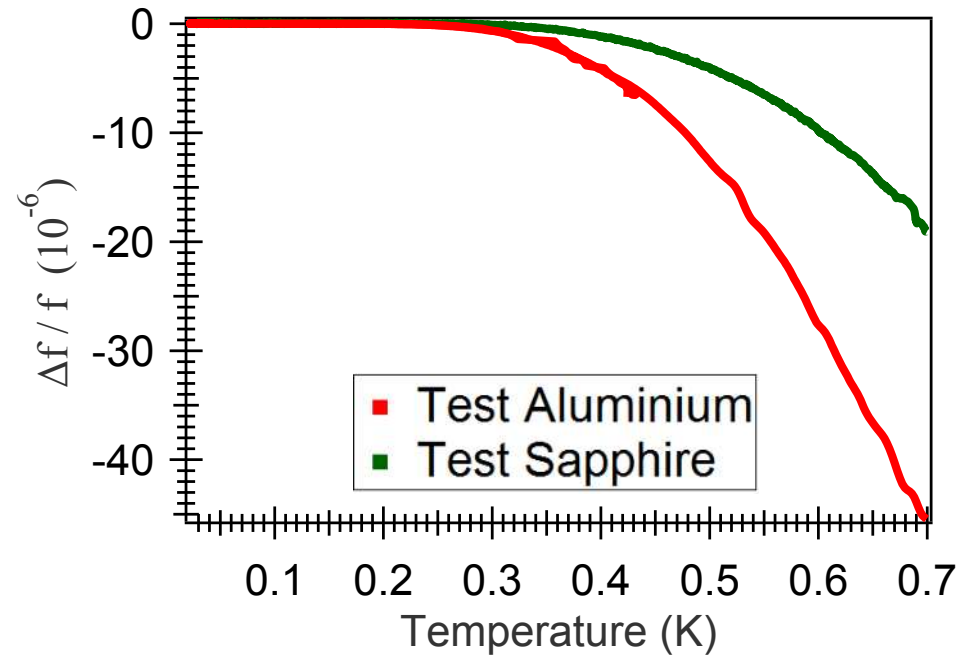
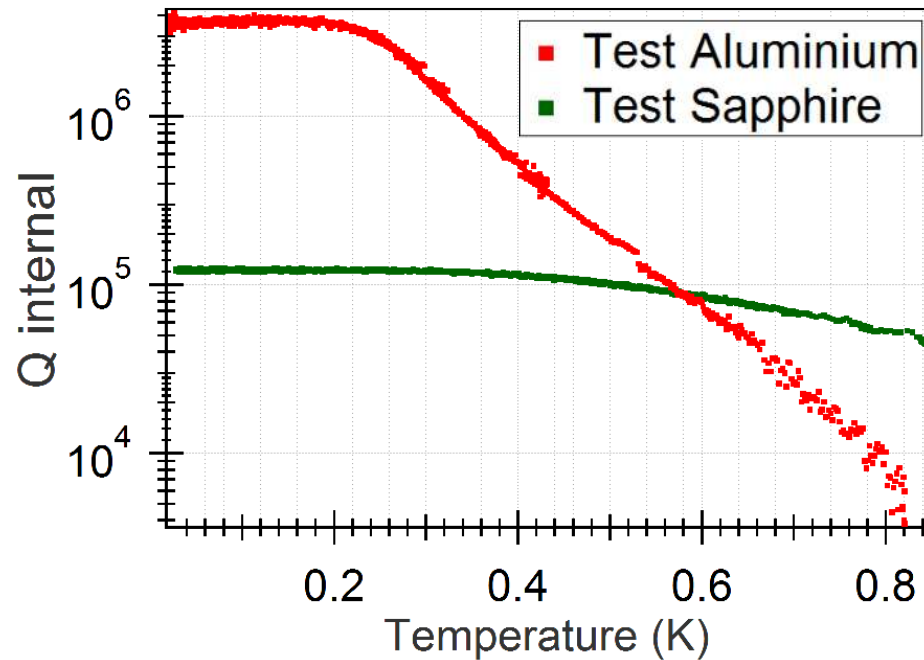
Normal Metal: $Q_s \sim 1$

Superconductor: $Q_s \sim 1/n_q$

Quasiparticles

Kinetic
Inductance

Loss Properties of Aluminum



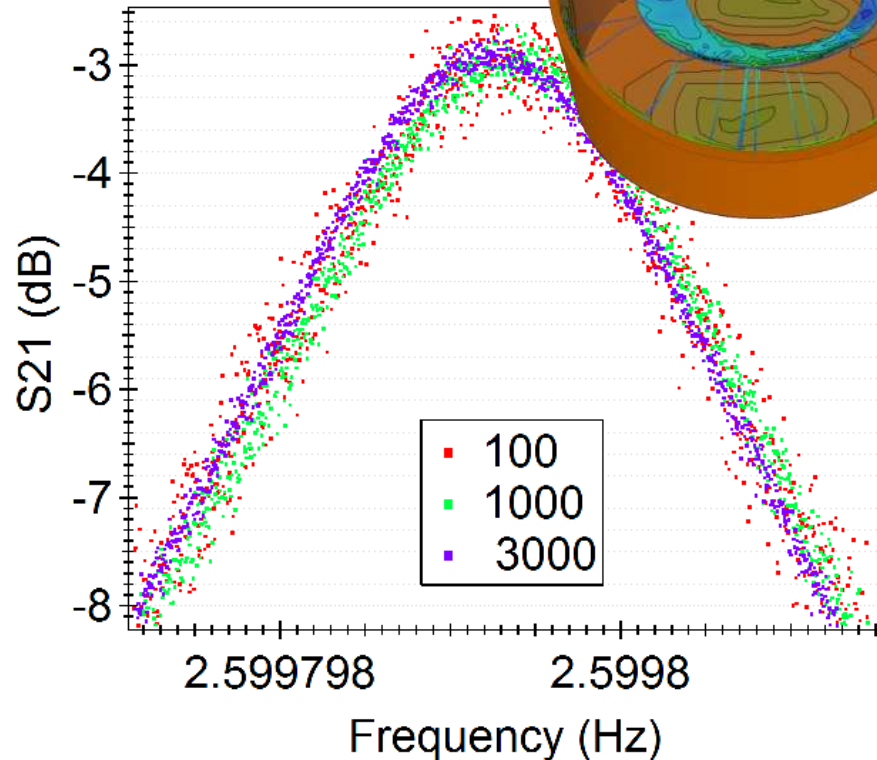
- Extracted upper bound of the surface square resistance of thin film aluminum:

$$R_{\square} \leq 250 \text{ n}\Omega$$
$$Q_s > 5000$$

See Matt Reagor's MLS for more details

Loss Properties of Sapphire

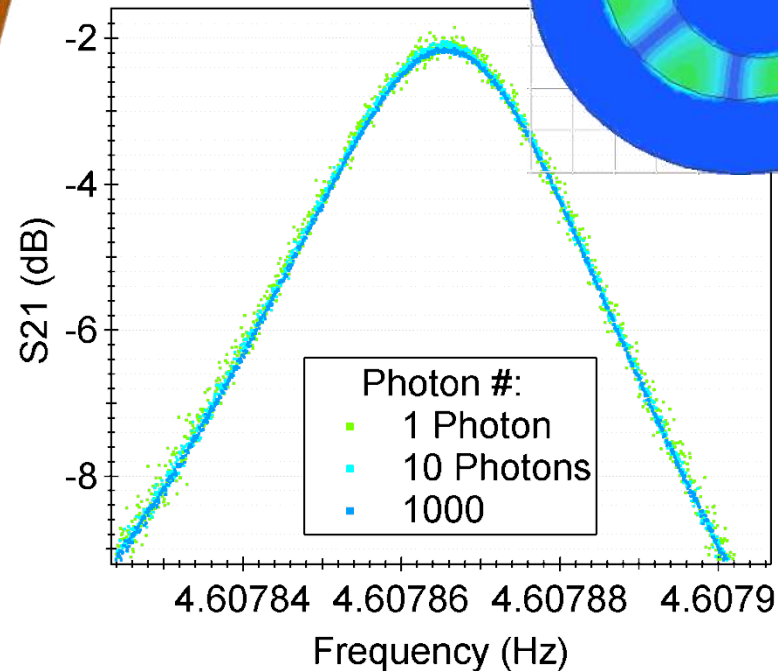
'Test Dielectric'



- HFSS participation ratio: 44%
- Common Perpendicular

$$Q_i = 1.6 \cdot 10^6$$
$$\tan \delta < 1.4 \cdot 10^{-6}$$

'Test Dielectric'

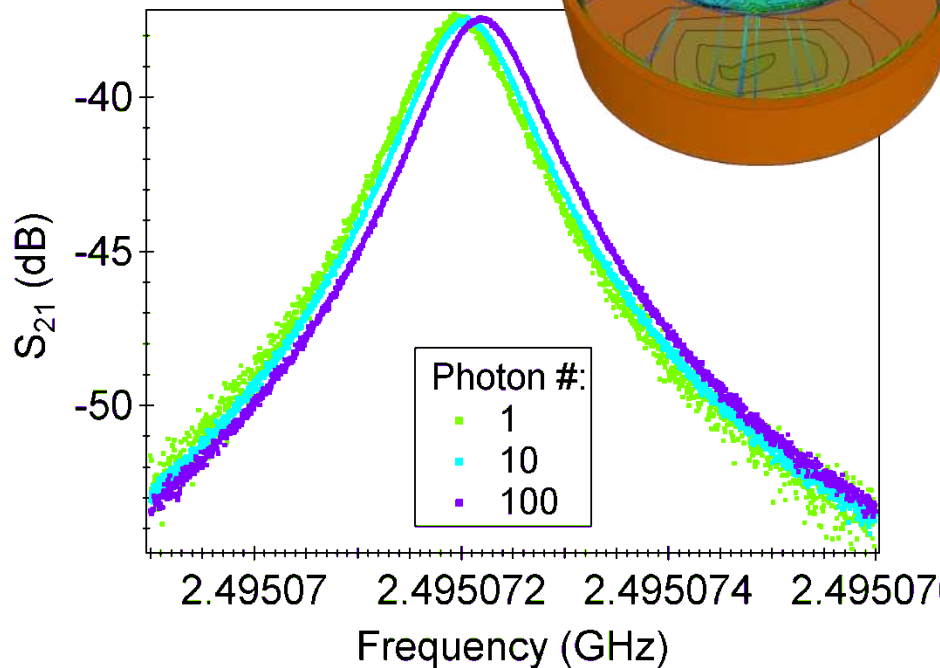


- HFSS participation ratio: 98%
- Differential 4th Harmonic

$$Q = 2.0 \cdot 10^5$$
$$\tan \delta < 5 \cdot 10^{-6}$$

Loss Properties of Sapphire

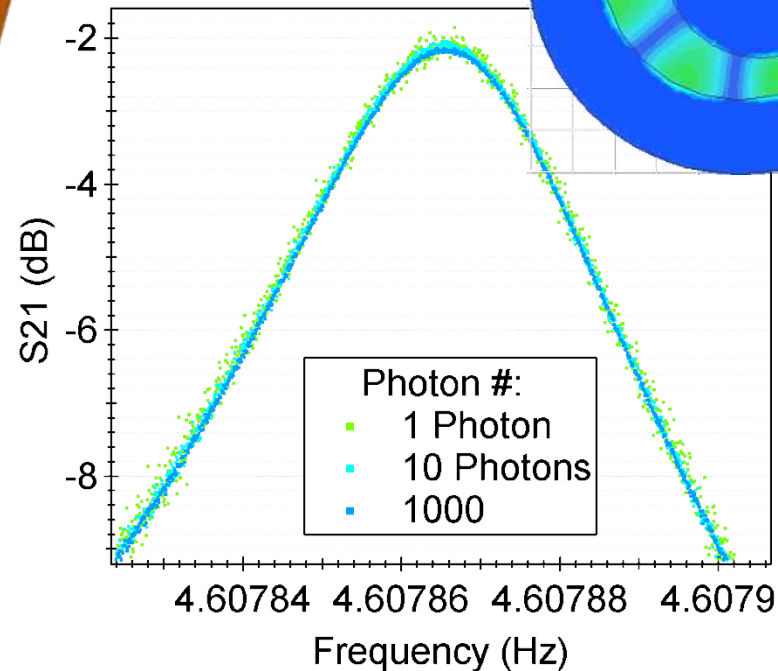
'Test Metal Setup'



- HFSS participation ratio: 44%
- Common Perpendicular

$$Q = 2.1 \cdot 10^6$$
$$\tan \delta < 1 \cdot 10^{-6}$$

'Test Dielectric'

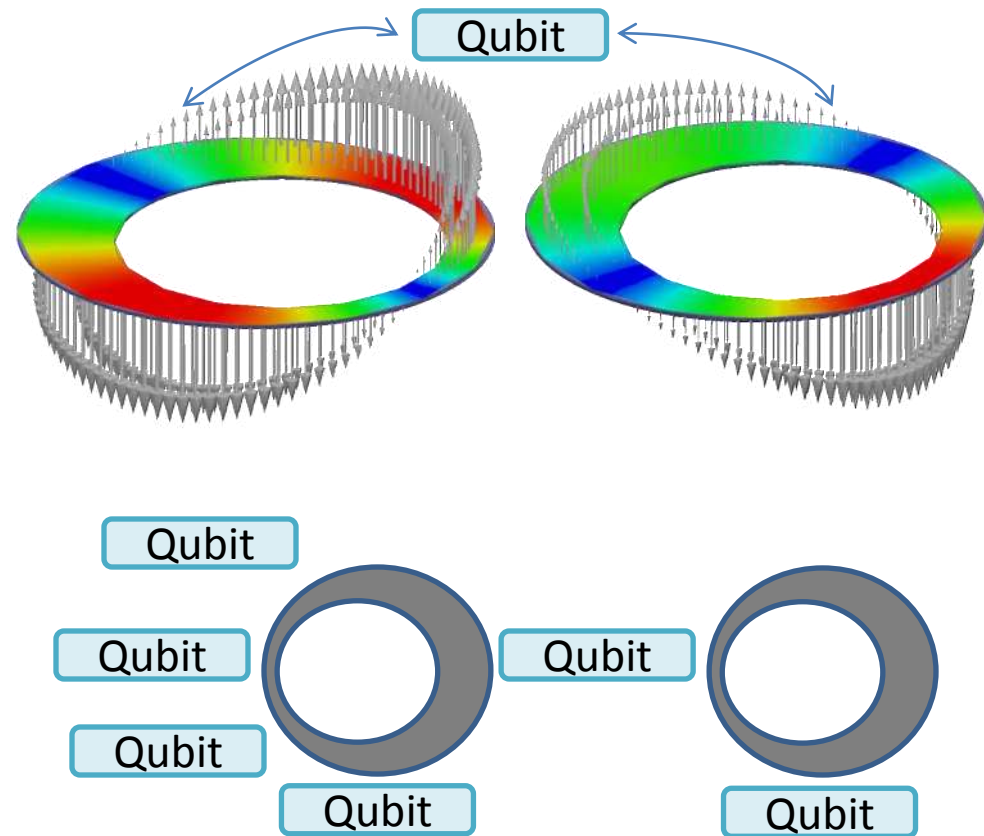


- HFSS participation ratio: 98%
- Differential 4th Harmonic

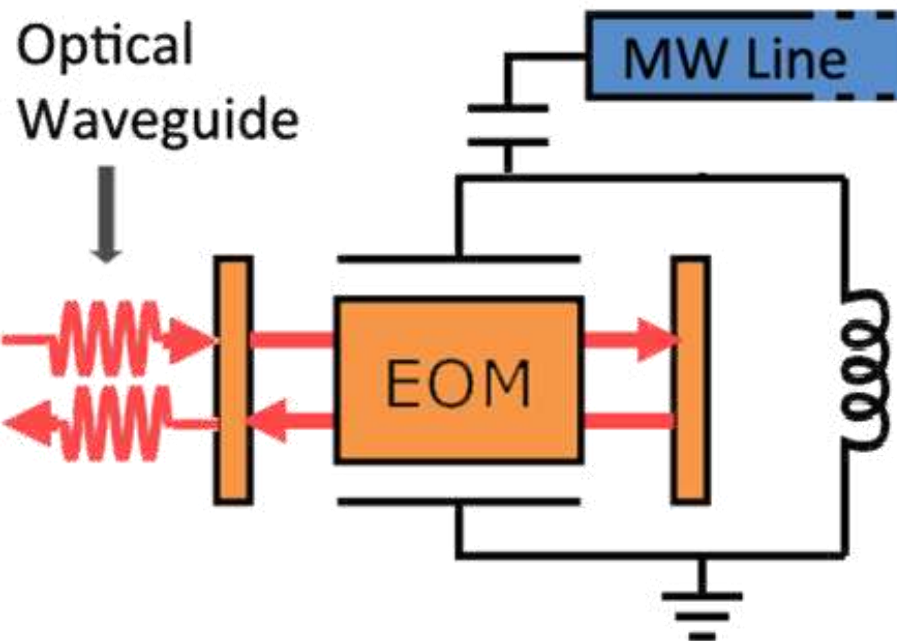
$$Q = 2.0 \cdot 10^5$$
$$\tan \delta < 5 \cdot 10^{-6}$$

Future Direction

- *Cryogenic material properties*
- *Inductive/Capacitive/Galvanic coupling to Fluxonium*
- Integration with multiple qubits and resonators in a 'Wafer-Scalable' Circuit-QED architecture.
- Quantum Bus



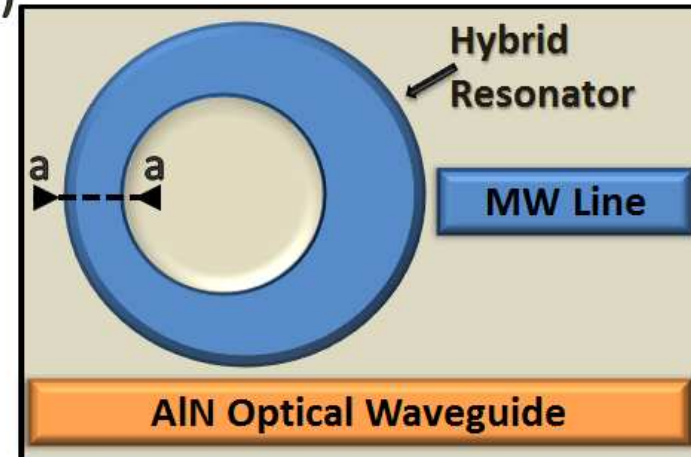
An Idea: EOM



Cross Section (a-a)



Top View



- $\mathcal{H}_C = -\hbar g(b + b^\dagger)a^\dagger a$
- $g \sim 1\text{kHz}$
- Entanglement
- Laser cooling
- Back-action evading optical measurements of mesoscopic devices

- $\chi^{(2)} \sim 4.7\text{ pm/V}$
(among the largest in integrated photonic)
- AlN WGMR $Q \sim 6E5$ at 1550 nm



QUANTRONICS LABORATORY

Department of Applied Physics

Yale University

